QUESTIONS

Q1
University Alumni Information Management System
50 marks

1/1 answered

Q1 University Alumni Information Management System

50 marks

A prestigious university maintains an alumni database containing graduates working across different industries worldwide. Alumni details are periodically updated through departmental coordinators, alumni associations, and online portals. As a result, duplicate alumni IDs, repeated email addresses, inconsistent company names, and multiple contact records are present. The university intends to develop a Java application that manages alumni records efficiently and generates analytical reports using Collection Framework classes.

Develop a Java application that performs the following operations:

a) Read Alumni ID, Name, Department, Graduation Year, Company Name, Designation, Email ID, and Country.

b) Use String Handling methods to standardize names and company names.

c) Store alumni records using an ArrayList.

d) Remove duplicate Alumni IDs using a HashSet.

e) Store Alumni IDs and Alumni Objects using a HashMap.

f) Traverse records using Iterator.

g) Use Generics in all collection implementations.

h) Generate department-wise alumni reports and compare the functionality of List, Set, and Map.

Your Code

```java
import java.util.*;

public class AlumniRecord {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        // g) Use Generics in all collection implementations
        // c) Store alumni records using an ArrayList
        List<Alumni> alumniList = new ArrayList<Alumni>();

        // d) Remove Duplicate Alumni IDs using a HashSet
        Set<Integer> idSet = new HashSet<Integer>();

        // e) Store Alumni IDs and Alumni Objects using a HashMap
        Map<Integer, Alumni> alumniMap = new HashMap<Integer, Alumni

        System.out.print("Enter number of alumni records: ");
        int n = Integer.parseInt(sc.nextLine().trim());

        for (int i = 0; i < n; i++) {
            System.out.println("\n--- Enter details for Alumni " + (

            // a) Read Alumni ID, Name, Department, Graduation Year,
            System.out.print("Alumni ID: ");
            int id = Integer.parseInt(sc.nextLine().trim());

            System.out.print("Name: ");
            String rawName = sc.nextLine();

            System.out.print("Department: ");
            String department = sc.nextLine().trim();

            System.out.print("Graduation Year: ");
            int gradYear = Integer.parseInt(sc.nextLine().trim());

            System.out.print("Company Name: ");
            String rawCompany = sc.nextLine();

            System.out.print("Designation: ");
            String designation = sc.nextLine().trim();

            System.out.print("Email ID: ");
            String email = sc.nextLine().trim();

            System.out.print("Country: ");
            String country = sc.nextLine().trim();

            // b) Use String Handling methods to standardize names an
            // trim() removes extra spaces, toLowerCase()/substring()
            String standardizedName = standardizeText(rawName);
            String standardizedCompany = standardizeText(rawCompany);

            // d) Remove Duplicate Alumni IDs using a HashSet
            if (idSet.contains(id)) {
                System.out.println("Duplicate Alumni ID " + id + " 
                continue;
            }
            idSet.add(id);

            Alumni alumni = new Alumni(id, standardizedName, departme
                                    standardizedCompany, designat

            // c) Store alumni records using an ArrayList
            alumniList.add(alumni);

            // e) Store Alumni IDs and Alumni Objects using a HashMap
            alumniMap.put(id, alumni);
        }

        // f) Traverse records using Iterator
        System.out.println("\n------ All Alumni Records (via Iterator)
        Iterator<Alumni> iterator = alumniList.iterator();
        while (iterator.hasNext()) {
            Alumni a = iterator.next();
            System.out.println(a);
        }

        // h) Generate department-wise alumni reports
        generateDepartmentWiseReport(alumniList);

        // h) Compare the functionality of List, Set, and Map
        compareCollections(alumniList, idSet, alumniMap);

        sc.close();
    }

    // b) String Handling method to standardize names/company names
    private static String standardizeText(String text) {
        text = text.trim();
        if (text.isEmpty()) return text;
        StringBuilder result = new StringBuilder();
        String[] words = text.toLowerCase().split("\\s+");
        for (String word : words) {
            if (word.length() > 0) {
                result.append(Character.toUpperCase(word.charAt(0)))
                        .append(word.substring(1))
                        .append(" ");
            }
        }
        return result.toString().trim();
    }

    // h) Generate department-wise alumni reports using a Map<String,
    private static void generateDepartmentWiseReport(List<Alumni> alu
        System.out.println("\n------ Department-wise Alumni Report ---

        // g) Generics used here as well
        Map<String, List<Alumni>> departmentMap = new HashMap<String,

        for (Alumni a : alumniList) {
            String dept = a.getDepartment();
            if (!departmentMap.containsKey(dept)) {
                departmentMap.put(dept, new ArrayList<Alumni>());
            }
```

Input

```
5
101
john kumar
Computer Science
```

Output

```
Enter number of alumni
records:
--- Enter details for
Alumni 1 ---
Alumni ID: Name:
Department: Graduation
```